

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_133119 | Base: main-v1.0

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Decision: GO_WITH_CONSTRAINTS | Confidence: 78%

CHANGE REQUEST

modifica la struttura in modo da renderla più lunga lungo l'asse y di un metro

CASCADE ISSUES IDENTIFIED

[CRITICAL] ISSUE-001 — Thermal gradient margin violation

Subsystem: thermal | Req: L1-THM-001 | Type: constraint_violation

Increasing the height of the structure will increase thermal gradients across the optical bench, potentially exceeding the ΔT limit of 15°C.

[MAJOR] ISSUE-002 — Missing shock test evidence

Subsystem: mechanical | Req: L1-MEC-003 | Type: ncr_required

The change may affect the structural dynamics, necessitating a new shock response analysis and testing to ensure compliance with launcher SRS ICD.

[MINOR] ISSUE-003 — Potential data processing latency increase

Subsystem: dataproc | Req: None | Type: reanalysis_required

The additional height may introduce more signal delay in the data processing pipeline, potentially affecting end-to-end latency requirements.

MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-32C Rev.1 §9 — Shock test per launcher SRS ICD

Reason: The structural change may affect the shock response characteristics, necessitating a new shock test.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	61%	WARN	ΔT 14.8°C — margin < 5%	—
mechanical	88%	OK	1st mode 112 Hz > 80 Hz req	—
structure	91%	OK	Mass 33.8 kg / budget 35 kg	—
solar	89%	OK	Power margin +18%	—
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

- L1-THM-001 — verification evidence invalidated, re-analysis required
- L1-MEC-003 — verification evidence invalidated, re-analysis required